Computer Networks Assignment Solution

Question 1:

Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

Answer:
I opened Wireshark and started capturing packets on my active network interface. Then, I refreshed the Flipkart homepage and stopped the capture. I examined the Protocol and packet header details to identify the TCP/IP layers.

TCP/IP layer	Example protocol or packet	How to identify it
Application	HTTP, DNS, TLS	Check the application protocol and message details.
Transport	TCP or UDP	Check source port, destination port, and TCP flags.
Internet	IPv4 or IPv6	Check source and destination IP addresses.
Network Access	Ethernet or Wi-Fi frame	Check MAC addresses and frame headers when available.

Conclusion: Wireshark helps us examine packet headers and understand how data travels through different TCP/IP layers.

Note: These are examples of what you may see, not a claim that all four were present in your actual capture.

Question 2:

Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

OSI Model (7 layers)	TCP/IP Model (4 layers)	Real-world example
7. Application	Application	Sending a WhatsApp message
6. Presentation	Application	Image formatting and encryption
5. Session	Application	Managing a communication session
4. Transport	Transport	TCP data delivery
3. Network	Internet	IP addressing and routing
2. Data Link	Network Access	Ethernet frames and MAC addresses
1. Physical	Network Access	Cables and Wi-Fi radio signals

Conclusion: The OSI model has seven layers, while the TCP/IP model has four layers. The TCP/IP model combines some of the OSI layers.

Question 3:

Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

Answer:

The TCP header contains fields that help devices communicate reliably.

TCP header field	Function
Source Port	Identifies the sending application's port.
Destination Port	Identifies the receiving application's port.
Sequence Number	Identifies the position of data in the byte stream.
Acknowledgment Number	Indicates the next byte expected from the other device.
Header Length	Specifies the size of the TCP header.
Flags	Control connection setup, acknowledgment, and termination (e.g., SYN, ACK, FIN).
Window Size	Indicates how much data the receiver can accept.
Checksum	Helps detect errors in the TCP header and data.
Urgent Pointer	Indicates urgent data when the URG flag is set.

Note: These are the standard TCP header fields. Label the fields that actually appear in the screenshot provided by your trainer.

Question 4:

Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

- **1. Application Layer:** I type a message in Instagram and press Send.
- **2. Transport Layer:** TCP or UDP carries data using transport ports. TCP splits data into segments when TCP is used.
- **3. Internet Layer:** IP adds source and destination IP addresses and routes packets.
- **4. Network Access Layer:** Data is placed into local network frames and transmitted over Wi-Fi or Ethernet.

Internet → Instagram Server

The server processes the request and handles delivery to the recipient.

Question 5: Identify One Packet in Wireshark

Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

Example Answer:

- **Protocol:** TLS
- **TCP/IP layer:** Application layer, for the purpose of this assignment.
- **How I identified it:** I checked the Protocol column in Wireshark and found TLS. The Info column showed Client Hello.
- **Real-world action:** This packet represents the start of a secure connection when my browser or app connects to a website.

Conclusion: Wireshark helps us identify network protocols, inspect packet headers, and understand how applications communicate over the internet.

Important: This is an example. Replace TLS and Client Hello with the details of a packet from your own capture if they differ.